Sharp Guarantees for Ensemble Voting
An Elementary Analysis of Dependence

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Abstract

Individual accuracies and pairwise correlations leave majority-vote accuracy undetermined, but they can still certify an improvement. After solving the assignment's marginal and three-voter cases, we give an elementary specialization of the classical discrete moment problem to every odd ensemble size. Integer quadratic inequalities and a planar construction provide the exact attainable accuracy interval. A one-line comparison of adjacent certificates reduces its evaluation to constant time. We then derive an explicit necessary-and-sufficient correlation threshold for voting to outperform each equally accurate member, with a three-point counterexample at the threshold. The improvement over the usual continuous moment guarantee separates exactly into a threshold correction and an integer-support correction. For accuracy 0.6 and pairwise independence, the first odd ensemble size with a strict guaranteed gain is 27, whereas the usual C-bound first certifies a gain at 37. These are population results, verified independently against finite linear programs.

1 Question, scope, and relationship to prior work

Let $K = 2m + 1 \geq 3$, $t = m + 1$, $Z_i = \mathbf{1}\{f_i(X) = Y\}$, and $p_i = \mathbb{P}(Z_i = 1)$. The accuracy of the binary majority vote is

$$S = \sum_{i=1}^{K} Z_i, \qquad A = \mathbb{P}(S \geq t). \tag{1}$$

The case $K = 1$ simply has $A = p_1$. Any joint law of the Z_i is realizable by fixed classifiers: take $X = Z$, $Y = 1$, and $f_i(X) = X_i$. Thus the extremal laws below respect the assignment's model.

We ask what marginal and pairwise data determine, and *when they guarantee that voting beats a member*. With equal accuracies this means $\inf A > p$, rather than improvement under just one favorable dependence structure.

What is already known. Kuncheva and Whitaker (2003) show that diversity alone does not determine accuracy; Kuncheva et al. (2003) study extreme patterns and dependence thresholds. Germain et al. (2015) derive the C-bound from margin moments, and Masegosa et al. (2020) study second-order oracle and PAC-Bayesian bounds. Sharp event-count inequalities, moment optimization, and count representations of exchangeable Bernoulli laws are classical (Boros and Prékopa, 1989; Bertsimas and Popescu, 2005; Fontana and Semeraro, 2021).

Contribution of this project. We do not claim the first sharp two-moment inequality. We give an elementary, constructive solution with constant-time evaluation, an explicit finite-K strict-benefit threshold, and a precise separation of two improvements over continuous bounds. This is a simplification and synthesis within the original problem. The proofs use quadratic inequalities and planar geometry; the repository records the literature comparison and reproducible checks.

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2 Part I: independence and unrestricted dependence

2.1 The independent benchmark

For iid Bernoulli(p) correctness indicators,

$$A_{\mathrm{ind}}(K,p) = \sum_{s=t}^{K} \binom{K}{s} p^s (1-p)^{K-s}. \tag{2}$$

Conditioning on the original K votes gives

$$A_{\mathrm{ind}}(K+2,p) - A_{\mathrm{ind}}(K,p) = \binom{K}{(K-1)/2} [p(1-p)]^{(K+1)/2} (2p-1). \tag{3}$$

Only counts m and $m+1$ can change the outcome after two votes are added. Hence accuracy strictly increases for $1/2 < p < 1$, strictly decreases for $0 < p < 1/2$, and stays $1/2$ at $p = 1/2$. Chebyshev's inequality applied to S/K shows $A_{\mathrm{ind}} \to 1$ for fixed $p > 1/2$, and $A_{\mathrm{ind}} \to 0$ for $p < 1/2$. For unequal independent accuracies, $\mathbb{P}(S = s)$ is the coefficient of z^s in $\prod_i (1 - p_i + p_i z)$; replacing the p_i by their mean is generally invalid.

2.2 Sharp marginal-only bounds

With equal marginals, the pointwise inequality $S \le m + (m+1)\mathbf{1}\{S \ge t\}$ gives

$$A_{-}^{\mathrm{marg}}(K,p) = \max\left\{0, \frac{Kp - m}{m+1}\right\}, \qquad A_{+}^{\mathrm{marg}}(K,p) = \min\left\{1, \frac{Kp}{m+1}\right\}. \tag{4}$$

For the lower bound when $Kp > m$, put probability $a = (Kp - m)/(m+1)$ on $S = K$ and $1 - a$ on $S = m$. Conditional on a count, choose its correct voters uniformly. Each marginal is p and accuracy is a. When $Kp \le m$, mix the adjacent integers around Kp, so the vote never wins. Complementing all correctness indicators proves and attains the upper bound; mixing endpoint laws fills the interval.

For $p > 1/2$, the lower bound equals

$$A_{-}^{\mathrm{marg}}(K,p) = 2p - 1 + \frac{2(1-p)}{K+1} \longrightarrow 2p - 1. \tag{5}$$

It decreases with odd K for $p < 1$, the opposite of independence. At $p = 0.6$, three voters range from 0.4 to 0.9, although the independent value is 0.648.

For unequal marginals, sort $p_{(1)} \ge \cdots \ge p_{(K)}$ and put $P_r = \sum_{i=1}^{r} p_{(i)}$. The complete marginal answer is

$$A_{-}^{\mathrm{marg}}(p_1, \ldots, p_K) = \max\left\{0, \max_{t \le r \le K} \frac{P_r - m}{r - m}\right\}, \qquad A_{+}^{\mathrm{marg}}(p) = 1 - A_{-}^{\mathrm{marg}}(1-p). \tag{6}$$

Any r voters have at most m successes on a losing vote, which proves every lower certificate. Appendix A gives the attaining construction. For $(0.9, 0.9, 0.1)$ the sharp minimum is 0.8; a mean-only calculation gives just 0.45.

Why weights alone cannot resolve ignorance. For nonnegative weights summing to one and strict weighted majority, the marginal-only minimax accuracy is $\max_i p_i$. Selecting the best classifier attains it. Conversely, $Z_i = \mathbf{1}\{U \le p_i\}$ with a common uniform U makes every rule fail whenever $U > \max_i p_i$. This explains why dependence information, rather than weight optimization alone, is necessary for a guaranteed ensemble gain.

3 Part II: the exact three-voter answer

Let $r_{ij} = \mathbb{E}Z_iZ_j$. For nondegenerate marginals the correlation fixes

$$r_{ij} = p_ip_j + \rho_{ij}\sqrt{p_i(1-p_i)p_j(1-p_j)}. \tag{7}$$

At $p_i = 0$ or 1, correlation is undefined; use joint moments directly. For $K = 3$ all eight probabilities are determined by $q = \mathbb{P}(111)$ and the first two moments:

Configuration	Probability
111	q
110, 101, 011	$r_{12} - q,\ r_{13} - q,\ r_{23} - q$
100	$p_1 - r_{12} - r_{13} + q$
010	$p_2 - r_{12} - r_{23} + q$
001	$p_3 - r_{13} - r_{23} + q$
000	$1 - \sum_i p_i + \sum_{i<j} r_{ij} - q$

Define

$$L = \max(0, r_{12} + r_{13} - p_1, r_{12} + r_{23} - p_2, r_{13} + r_{23} - p_3), \tag{8}$$
$$U = \min(r_{12}, r_{13}, r_{23}, 1 - \textstyle\sum_i p_i + \sum_{i<j} r_{ij}). \tag{9}$$

Nonnegativity is exactly $L \le q \le U$, so the moments are feasible iff $L \le U$. Since $A = \sum_{i<j} r_{ij} - 2q$, the complete attainable interval is

$$\boxed{\sum_{i<j} r_{ij} - 2U \ \le A \le \sum_{i<j} r_{ij} - 2L.} \tag{10}$$

These are exact distributions, not just necessary inequalities.

3.1 Equal accuracies and equal correlations

For $p_i = p$, $\rho_{ij} = \rho$, put $r = p^2 + \rho p(1-p)$. Then

$$L = \max(0, 2r - p), \quad U = \min(r, 1 - 3p + 3r), \quad A_{\min} = 3r - 2U, \quad A_{\max} = 3r - 2L. \tag{11}$$

The feasible interval is $\max(0, p - 1/3, 2p - 1) \le r \le p$. For $Q_s = \mathbb{P}(S = s)$, the count law is

$$(Q_0, Q_1, Q_2, Q_3) = (1 - 3p + 3r - q,\ 3p - 6r + 3q,\ 3r - 3q,\ q). \tag{12}$$

At $p = 0.6$, $\rho = 0$, each pair is independent, yet:

Law	Q_0	Q_1	Q_2	Q_3	A
Worst admissible	0	.48	.24	.28	.52
Joint independence	.064	.288	.432	.216	.648
Best admissible	.16	0	.72	.12	.84

Uniform allocation within each count gives the same p and r in all rows. The unobserved triple success probability changes the vote. At $p = 1/2, \rho = 0$ the interval is $[1/4, 3/4]$. Also, Bernoulli feasibility is stronger than positive semidefiniteness of the correlation matrix: for $p = 1/2$ it requires $\rho \ge -1/3$, whereas an equicorrelation matrix permits $\rho = -1/2$.

4 A sharp integer-moment solution for every odd K

4.1 Symmetrization removes exponential enumeration

Assume all $p_i = p$ and all $r_{ij} = r$. Uniformly permuting classifier indices preserves these constraints and A. Conversely, any count law on $\{0, \dots, K\}$, followed by a uniformly chosen subset of the specified size, has

$$\mu := \mathbb{E}S = Kp, \qquad \nu := \mathbb{E}S(S-1) = K(K-1)r. \tag{13}$$

Thus this reduction is exact; it does not assume the original classifiers were exchangeable. Feasible moments form the polygon with vertices $v_s = (s, s(s-1))$. If $b = \lfloor \mu \rfloor$, feasibility is exactly

$$0 \le \mu \le K, \qquad 2b\mu - b(b+1) \le \nu \le (K-1)\mu. \tag{14}$$

Indeed, adjacent counts minimize the second factorial moment at fixed mean, and counts $0, K$ maximize it. Mixing those laws fills the interval. This is finite exchangeability, with no iid-mixture assumption that would incorrectly exclude negative correlation.

4.2 Quadratic certificates and the exact bound

Write $D_j = (j-m)(j+1-m)$ and define

$$B_K(\mu, \nu) = \max\left\{0, \frac{\nu - (m-1)\mu}{K(K-m)}, \max_{m+1 \le j \le K-1}\left[1 - \frac{\nu - 2j\mu + j(j+1)}{D_j}\right]\right\}. \tag{15}$$

Theorem 1 (Sharp count bound)**.** *For every feasible pair (μ, ν), the complete attainable accuracy interval is*

$$B_K(\mu, \nu) \ \le A \le \ 1 - B_K\big(K - \mu, K(K-1) - 2(K-1)\mu + \nu\big). \tag{16}$$

Each endpoint is attained by a count law supported on at most three integers, followed by uniform allocation of successes to classifiers.

Proof of validity. For every integer $s \in \{0, \dots, K\}$, the following functions lie below $\mathbf{1}\{s \ge m+1\}$:

$$0, \qquad h(s) = \frac{s(s-m)}{K(K-m)}, \qquad g_j(s) = 1 - \frac{(s-j)(s-j-1)}{D_j}. \tag{17}$$

For $s \le m$, $h(s) \le 0$, and $(s-j)(s-j-1) \ge D_j$ gives $g_j(s) \le 0$. For $s \ge m+1$, $s(s-m) \le K(K-m)$, while the product of two consecutive integers is nonnegative, giving $g_j(s) \le 1$. Taking expectations proves (15). Replacing S by $K - S$ proves the upper bound. The attaining construction follows on the next page. □

This proof shows exactly where discreteness matters: a quadratic can be nonnegative on every integer while negative between consecutive integers. A continuous moment bound discards that useful restriction. For $K = 3$, (15) reduces to $\max(0, r, 6p - 3r - 2)$, precisely the lower endpoint in (11). Unlike a 2^K-cell optimization, the certificate family grows only linearly in K; it can be evaluated even faster.

4.3 An elementary attaining construction

Draw a fan from v_m in the moment polygon. Its pieces are the lower-count polygon $\mathrm{conv}(v_0, \ldots, v_m)$, the bridge triangle $\mathrm{conv}(v_0, v_m, v_K)$, and the triangles $\mathrm{conv}(v_m, v_j, v_{j+1})$, $m+1 \le j \le K-1$. Every feasible moment point lies in one of these pieces. Its convex coefficients specify a count law:

- In the lower-count polygon, all counts are at most m, so $A = 0$. A fan triangulation inside it needs at most three counts.
- In the bridge triangle, h equals the event indicator at $0, m, K$; the law attains $\mathbb{E}h$.
- In a remaining triangle, g_j equals the indicator at $m, j, j+1$; the law attains $\mathbb{E}g_j$.

Because all certificates are lower bounds, the matching certificate must equal their maximum. This completes the proof of Theorem 1, including boundaries and zero weights. Mixtures of endpoint laws give every intermediate accuracy. Figure 1 visualizes the construction.

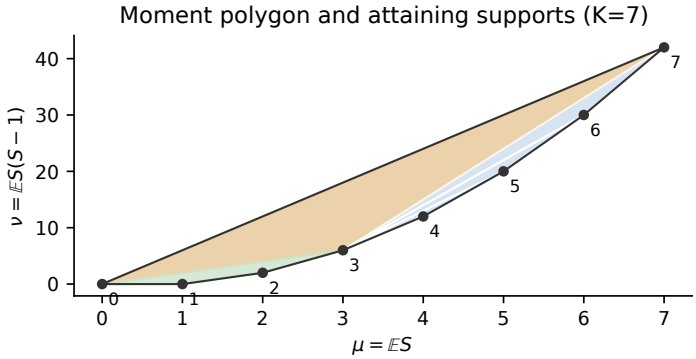


Figure 1: The feasible moment polygon. Green uses only losing counts; orange attains h; blue triangles attain successive g_j. Vertex labels are counts s.

4.4 Constant-time evaluation by comparing adjacent certificates

Put $d = \mu - m$, $v = \mathbb{E}(S-m)^2 = \nu - (2m-1)\mu + m^2$, and $j = m + a$. Then

$$G_a := \mathbb{E}g_{m+a}(S) = \frac{(2a+1)d - v}{a(a+1)}, \quad 1 \le a \le m, \qquad G_a - G_{a+1} = \frac{2[(a+1)d - v]}{a(a+1)(a+2)}. \tag{18}$$

If $d \le 0$, every $G_a \le 0$. If $d > 0$, the sequence increases until $a+1$ crosses v/d and then decreases. Consequently a maximizing index is

$$a_* = \min\{m, \max\{1, \lfloor v/d \rfloor\}\}. \tag{19}$$

Thus B_K requires only the comparison of zero, $\mathbb{E}h$, and G_{a_*}; ties at integer ratios are harmless. This is $O(1)$ arithmetic once μ, ν are known. Reading arbitrary pairwise inputs would itself take $O(K^2)$ time; we make no constant-time claim for that task. On an adjacent triangle, the explicit probabilities are

$$Q_{m+a} = \frac{(a+1)d - v}{a}, \quad Q_{m+a+1} = \frac{v - ad}{a+1}, \quad Q_m = 1 - Q_{m+a} - Q_{m+a+1}. \tag{20}$$

Use these only on that triangle; the bridge has a different attaining law.

5 Exactly when is voting guaranteed to beat its members?

Now fix $1/2 < p < 1$. Every member has accuracy p, so a strict guaranteed gain means $A_{\min} > p$. Let

$$x = \frac{Kp - (1-p)m}{p}, \quad j = \lfloor x \rfloor, \quad \theta = x - j, \tag{21}$$

$$\nu_* = (1-p)m(m-1) + p[2jx - j(j+1)], \tag{22}$$

$$\rho_* = \frac{\nu_*/[K(K-1)] - p^2}{p(1-p)}. \tag{23}$$

Here $m + 1 < x < K$, so j is a valid certificate index.

Theorem 2 (Exact strict-benefit threshold)**.** *For every feasible common correlation ρ,*

$$\boxed{A_{\min}(K, p, \rho) > p \quad \Longleftrightarrow \quad \rho < \rho_*.} \tag{24}$$

At equality, $A_{\min} = p$. For every $\rho \geq \rho_*$ in the feasible range, there is an admissible law whose majority accuracy is exactly p.

Proof. Since $\nu \leq K(K-1)p$, the bridge certificate satisfies $\mathbb{E}h \leq p$. Thus a strict gain occurs iff at least one g_ℓ has expectation greater than p. Rearranging this inequality gives $\nu < T_\ell$, where

$$\begin{aligned} T_\ell &= 2\ell Kp + (1-p)(\ell - m)(\ell + 1 - m) - \ell(\ell+1) \\ &= (1-p)m(m-1) + p[2\ell x - \ell(\ell+1)]. \end{aligned} \tag{25}$$

This is a concave quadratic in the integer ℓ. Its maximum is at $\ell = \lfloor x \rfloor$, with a possible adjacent tie when x is integral. The maximum is exactly ν_*. Since ν increases strictly with ρ, this proves (24).

For a direct boundary construction, put probabilities

$$(1-p,\ p(1-\theta),\ p\theta) \quad \text{on counts} \quad (m, j, j+1). \tag{26}$$

Its mean is Kp, factorial second moment is ν_*, and accuracy is p. Uniform allocation realizes all the specified one- and two-voter moments. The same certificate gives $A \geq p$, proving equality at the threshold. Finally, mix (26) with the synchronized law $\mathbb{P}(S = K) = p$, $\mathbb{P}(S = 0) = 1 - p$. This fills every moment from ν_* to $K(K-1)p$ while preserving accuracy p, proving the last assertion. □

For $K = 3$, this simplifies to $\rho_* = 1 - 2/(3p)$. In particular $p = 0.6$ needs $\rho < -1/9$ for a strict guaranteed gain: zero or slightly negative correlation does not suffice. This resolves the question quantitatively rather than treating the sign of correlation as a decision rule. For larger K, only one floor operation is needed, and the explicit counterexample explains why the condition cannot be weakened.

6 What is gained over continuous moment bounds?

Let $\sigma^2 = \mathrm{Var}(S) = Kp(1-p)[1+(K-1)\rho]$ and $d = Kp - m > 0$. The ordinary C-bound applied to $M = 2S/K - 1$ guarantees

$$A \geq C_K = \frac{(2p-1)^2}{(2p-1)^2 + 4\sigma^2/K^2}. \tag{27}$$

A fairer elementary baseline first uses the correct failure threshold $S \leq m$. Cantelli then gives

$$A \geq C_K^{\mathrm{shift}} = \frac{d^2}{d^2 + \sigma^2}. \tag{28}$$

Since Theorem 1 is exact for the specified integer support, its lower endpoint is at least both bounds. The latter can still be loose when Cantelli's attaining second support point is nonintegral or outside $[0, K]$.

The boundary law (26) has variance $(1-p)d^2/p + p\theta(1-\theta)$. Consequently

$$\rho_* = \frac{d^2}{K(K-1)p^2} - \frac{1}{K-1} + \frac{\theta(1-\theta)}{K(K-1)(1-p)}. \tag{29}$$

The first two terms are exactly the strict-benefit threshold certified by shifted Cantelli. The last, nonnegative term is the additional integer-support correction. It vanishes when x is integral. For the original C-bound threshold $\rho_C = [K(2p-1)^2/(4p^2) - 1]/(K-1)$, the full difference is

$$\rho_* - \rho_C = \underbrace{\frac{p - 1/2 + 1/(4K)}{(K-1)p^2}}_{\text{correct failure threshold}} + \underbrace{\frac{\theta(1-\theta)}{K(K-1)(1-p)}}_{\text{integer support}}. \tag{30}$$

Both comparisons concern population guarantees, not finite-sample estimation or learned weights. For fixed $p \in (1/2, 1)$,

$$\lim_{K\to\infty,\ K \text{ odd}} \rho_*(K,p) = \left(1 - \frac{1}{2p}\right)^2. \tag{31}$$

This is a limit of finite-ensemble thresholds, not an assumption that the extremal laws for different K form one infinite exchangeable sequence.

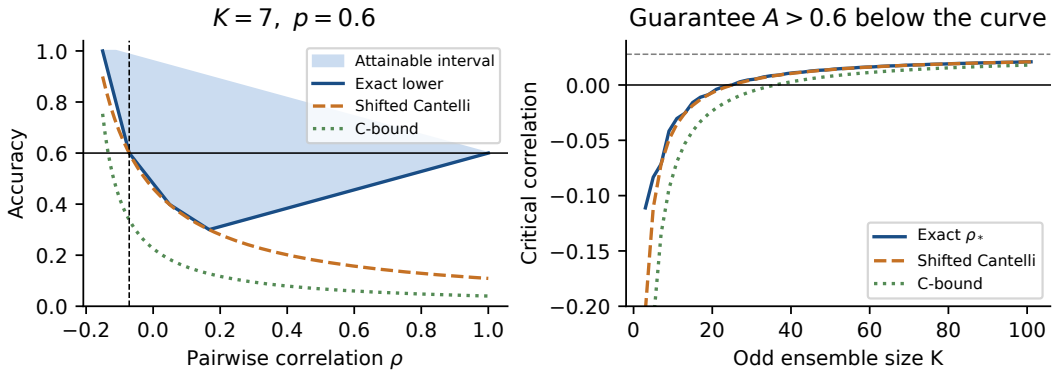


Figure 2: Exact attainable intervals and guarantee thresholds at $p = 0.6$. Left: accuracy need not decrease monotonically with correlation. Right: a strict gain is certified below the respective curve, within the feasible correlation range. The vertical range is cropped to show the finite-K comparison; the gray horizontal line is the limiting threshold.

7 Reproducible checks, a concrete consequence, and limitations

7.1 Checks designed to falsify the formulas

The repository includes a notebook, pinned Python dependencies, and an executable script using seed 20261013. It verifies every quadratic inequality on its integer support, constructs attaining laws, and compares the analytic endpoints with independent optimization. The count LP minimizes/maximizes $\sum_{s>m} Q_s$ subject to normalization and the two moments. For $K \le 7$, a separate 2^K-configuration LP constrains *every* p_i and r_{ij}. Its objectives and constraints do not use the analytic bounds.

The completed run checks 702 count-moment instances (both extrema), 41 full binary instances (both extrema), 720 threshold instances, 2,576 individual certificate values, and 36 invalid moment pairs. Cases include point masses, sparse and dense random laws, $p = 0, 1$, polygon boundaries, and integral x. The largest absolute count-LP discrepancy is below 3×10^{-14}. The constant-time formula is additionally checked against the full certificate maximum. These computations detect implementation errors; the proofs establish the claims beyond the checked cases.

7.2 Pairwise independence is eventually sufficient, but not immediately

For $p = 0.6$, $\rho = 0$, Table 1 compares attainable bounds with joint independence and the two continuous guarantees. The initial drop from $K = 3$ to $K = 5$ concerns the worst laws at each size, not an assertion about extending one fixed joint law.

Table 1: Population accuracy at $p = 0.6$, $\rho = 0$; all numbers follow from formulas.

K	Exact min	Exact max	Independent	C-bound	Shifted Cantelli
3	0.520000	0.840000	0.648000	0.111111	0.470588
5	0.466667	0.920000	0.682560	0.172414	0.454545
7	0.480000	0.960000	0.710208	0.225806	0.461538
23	0.592000	1.000000	0.836357	0.489362	0.586826
25	0.600000	1.000000	0.846232	0.510204	0.600000
27	0.616000	1.000000	0.855348	0.529412	0.612440
31	0.640000	1.000000	0.871618	0.563636	0.635294

Corollary 1. *With common accuracy $3/5$ and pairwise independence, majority voting has a strict guaranteed gain iff odd $K \ge 27$.*

Indeed, for $K = 2m + 1$, substituting $p = 3/5$ into (29) yields

$$\rho_* = \frac{m - 12 + (5/m)\mathbf{1}\{3 \nmid m\}}{18(2m + 1)}. \tag{32}$$

It is negative for $1 \le m \le 11$, zero at $m = 12$, and positive for $m \ge 13$. Theorem 2 proves the claim. Shifted Cantelli also certifies the first strict gain at 27 here because the boundary at 25 has integral x. The ordinary C-bound is $K/(K + 24)$ and requires $K > 36$. We therefore do not attribute this entire improvement to the lattice correction.

Limitations and conclusion. For unequal specified pairwise moments, the general configuration LP remains exact (Appendix B); aggregate count moments alone may lose sharpness. Estimated moments need uncertainty regions, and no finite-sample confidence claim is made here. The main insight is constructive: pairwise information need not identify accuracy to certify a gain, and finite integer support can make that certification sharper and simpler than a generic moment calculation.

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A Attainment for heterogeneous marginals

We prove the lower formula (6); complementing correctness indicators gives its upper counterpart. The following proof is retained from the initial report to keep all assignment cases fully justified.

Proof. Write $E = \{S \geq t\}$ and $a = \mathbb{P}(E)$. For a subset I of $r \geq t$ voters, at most $t-1$ of them are correct on E^c and at most r on E. Consequently,

$$\sum_{i \in I} p_i = \mathbb{E} \sum_{i \in I} Z_i \leq (t-1)(1-a) + ra, \tag{33}$$

so $a \geq [\sum_{i \in I} p_i - (t-1)]/(r-t+1)$. For fixed r, the largest sum uses the r largest marginals; this proves the lower bound.

To prove attainment, set a equal to the right-hand side of (6) and split each marginal as

$$p_i = x_i + y_i, \quad x_i = (p_i - a)_+, \quad y_i = \min(p_i, a). \tag{34}$$

We first show $\sum_i x_i \leq (t-1)(1-a)$. Since $\sum_i (p_i - a)_+ = \max_{I \subseteq [K]} \sum_{i \in I} (p_i - a)$, consider a maximizing subset of size r. If $r \leq t-1$, then its sum is at most $r(1-a) \leq (t-1)(1-a)$. If $r \geq t$, the definition of a and (33) give the same inequality.

We also need $\sum_i y_i \geq ta$. This condition deserves attention: merely saturating all the upper constraints on E^c does not automatically make the proposed E viable. Let a_0 be the probability of E under the independent Bernoulli coupling with marginals p_i. The already proved lower bound gives $a \leq a_0$. Under that coupling, $\sum_i \min(p_i, a_0) \geq \sum_i \mathbb{P}(Z_i = 1, E) = \mathbb{E}[S \mathbf{1}_E] \geq t a_0$. The function $h(u) = \sum_i \min(p_i, u) - tu$ is concave and has $h(0) = 0$. Therefore $h(a) \geq 0$ for $0 \leq a \leq a_0$, as required.

For $0 < a < 1$, the vectors $x/(1-a)$ and y/a lie in

$$C_0 = \{v \in [0,1]^K : \sum_i v_i \leq t-1\},$$

$$C_1 = \{v \in [0,1]^K : \sum_i v_i \geq t\},$$

respectively. Both polytopes are convex hulls of their 0–1 vectors: at a vertex, two fractional coordinates could be perturbed in opposite directions if the sum constraint were active, while a lone

fractional coordinate could be perturbed if it were inactive; one fractional coordinate cannot make an active integer-valued sum. Hence we can realize $x/(1-a)$ using outcomes with at most $t-1$ correct voters, and y/a using outcomes with at least t correct voters. Mix these laws with probabilities $1-a$ and a. Equation (34) gives every marginal p_i, and the event probability is exactly a. The cases $a=0$ or 1 follow by using only the corresponding cardinality polytope. □

B General pairwise constraints and exact verification LPs

For arbitrary p_i, r_{ij}, index configurations by $B \subseteq [K]$ and let w_B be the probability that exactly B is correct. The exact minimum/maximum is

$$\begin{aligned} \min / \max \quad & \sum_{|B|\geq t} w_B, \\ \text{subject to} \quad & \sum_{B} w_B = 1, \quad w_B \geq 0, \\ & \sum_{B\ni i} w_B = p_i, \quad \sum_{B\supseteq\{i,j\}} w_B = r_{ij} \quad (i<j). \end{aligned} \tag{35}$$

These constraints are necessary and sufficient because the variables are the full joint law. This is the full binary LP used in verification. In the symmetric case, the independent count LP is

$$\min / \max \sum_{s=m+1}^{K} Q_s, \qquad Q_s \geq 0, \quad \sum_s Q_s = 1, \quad \sum_s sQ_s = \mu, \quad \sum_s s(s-1)Q_s = \nu.$$

The proof in the main text derives its answer without invoking a solver. Symmetric moments are essential for the converse construction to satisfy each specified pair; for heterogeneous pair data the count bound is still valid but need not be exact.

C Reproduction and submission statements

From the repository root run `python experiments/verify.py` and `python experiments/plot.py`. The notebook `project.ipynb` calls these same implementations rather than maintaining a second copy. Results are saved to `results/verification.json` and `results/comparison.csv`; figures are generated as PDF and PNG. Dependencies are pinned in `requirements.txt`. The README gives PDF build and submission-renaming instructions. Reading notes in `notes/literature.md` distinguish actual full-text access from metadata-only references; `notes/derivation.md` records the elementary arguments.

Individual contributions — complete before submission. Replace the generic team authorship with the actual team members and add one accurate contribution statement per member. No contributions or names have been invented. The assigned GroupID is also needed for the final submission filenames.

AI use. OpenAI Codex assisted with literature retrieval and reading notes, mathematical exploration, the elementary certificate and threshold derivations, implementation of reproducible SciPy checks, and drafting the report and presentation. The team must understand and independently verify all results and update this statement to reflect its final work.